

ZAIN ABBAS

AI/NLP Engineer | Automation & Full-Stack Development | Project & Business Management

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Summary

Engineer and founder-operator spanning AI/NLP research, full-stack and mobile development, automation engineering, and business leadership. An **IEEE Access** paper on multilingual hate-speech detection hit **95.7% accuracy**; production work has shipped web, mobile, and cloud platforms across **200+ live operations**. Currently directs two SECP-registered mining companies and a government-approved civic-tech venture.

Experience

CEO & Director | Zircon Mines & Durr Mines Pvt Ltd

Jan 2026 – Present

- Serve as CEO & Director of Zircon Mines Pvt Ltd and Director of Durr Mines & Minerals Pvt Ltd, leading two SECP-registered mining enterprises with government exploration licenses and leases across **six sites** in Gilgit-Baltistan spanning copper, gold, lithium, antimony, and nephrite.
- Own regulatory documentation and compliance for both companies, and lead investor relations with Chinese and other international stakeholders alongside community and public relations across operating regions.

AI Automation Engineer | Horizon Bee Tech

Oct 2025 – Present

- Migrated the camp-management platform's web app off a no-code builder (Lovable) and merged it with the React Native mobile app on one Supabase PostgreSQL backend. Consolidated records for **200+ operations** and cut administrative processing time by an estimated **60%**.
- Recruited and trained a 2-person engineering team, then built onboarding practices that handed technical ownership to non-technical field staff.
- Built a computer-vision football player-tracking system in Python that generated real-time analytics for coaching and resource-allocation decisions.
- Integrated the Reachinbox autoresponder and SignalWire SMS pipeline into stakeholder communications, cutting manual outreach effort by an estimated **70%**.

Founder & Research Lead | Zircon Logics

Jul 2025 – Present

- Founded Zircon Logics to commercialize a year of NLP research into a real-time, multilingual hate-speech monitoring dashboard for public-sector deployment, serving the model to the dashboard through a FastAPI service. The research produced an **IEEE Access** publication at **95.7% accuracy**.
- Pitched the solution to senior Gilgit-Baltistan government officials and secured formal approval for deployment, with rollout pending standard administrative processing. Wrote the RFP and managed the proposed infrastructure budget.
- Led a cross-disciplinary team spanning research, product development, and client engagement.

Software Engineering Intern | Horizon Bee Tech

Jun 2025 – Sep 2025

- Developed Mine Explorer, a Next.js/TypeScript web platform for virtual mine-site visits and centralized operational data across three mining companies, including an investor-relations desk for external stakeholders. Deployed on Vercel.
- Built Aqua Gauge, a rainfall-prediction model in Python using KNN and linear regression, for climate-resilience planning in remote communities.

IT Intern | Government IT Department, Gilgit-Baltistan

Jun 2024 – Sep 2024

- Developed a Tourism Information System portal with real-time booking and ratings features for tourism operators and remote communities.
- Developed a Student Management System for remote schools, digitizing enrollment records and automating grading workflows.

Education

BS Computer and Information Science | PIEAS (Pakistan Institute of Engineering and Applied Sciences) 04-09-2021 – 09-09-2025

Thesis: Fine-tuned XLM-RoBERTa for multilingual hate-speech detection (English, Urdu, Roman Urdu) at 95.7% accuracy, with LIME and SHAP explainable-AI integration for civic transparency.

Publications

Z. Abbas et al., "MUST: An Explainable AI-Based Framework for Multilingual Hate Speech Detection," *IEEE Access*, 2025. DOI: 10.1109/ACCESS.2025.3629527. Fine-tuned XLM-RoBERTa for multilingual hate-speech classification (English, Urdu, Roman Urdu) with LIME and SHAP explainability for content-moderation transparency.

Technical & Business Skills

AI & Machine Learning: Python, NLP, XLM-RoBERTa, mBERT, LSTM, PyTorch, TensorFlow, LIME, SHAP, Computer Vision, KNN, Linear Regression

Automation & Integration: FastAPI, n8n, Zapier, Make, Reachinbox, SignalWire, REST APIs, Workflow Automation

Full-Stack & Mobile: React Native, Next.js, TypeScript, JavaScript, Tailwind CSS, Node.js, Supabase (PostgreSQL), Database Design, Git, Docker, Vercel, C++

Business & Project Management: Agile Project Management, Stakeholder Coordination, Donor Relations, Investor Relations, Regulatory Compliance, RFP Writing, Budget Management

Leadership & Domain: Cross-functional Team Leadership, Community & Public Relations, Civic Tech, Mining & Natural Resources, NGO Operations, Public-Sector Deployment

Certifications

AI in Cyber Security | Macquarie University

Generative AI and Prompt Engineering | Edureka

Generative AI Tools and Automation | Edureka

Generative AI for Workflow Automation | Edureka

Project Initiation | Coursera

Project Planning | Coursera

Project Execution | Coursera

Agile Project Management | Coursera

Capstone: Applying Project Management in Real Life | Coursera

Leadership & Activities

- **Director, Media & Communications** | PIEAS Media Club (2024–2025): Led cross-functional media operations for large institutional events and managed a team of student volunteers.
- **Media Head** | Google Developer Student Clubs (2023–2024): Grew workshop attendance and expanded the community's digital reach through outreach and media strategy.
- **Media Director** | TEDx PIEAS (2023–2024): Owned outreach, media strategy, and communications for the event's "The Other Side" theme.
- **Team Captain** | PIEAS Football & Volleyball (2021–2025): Managed team logistics and performance at national inter-university competitions.

References

Dr. Ijaz Hussain, Principal Scientist & Advisor, PIEAS | ijazhussain@pieas.edu.pk

Dr. Abdul Majid, Chief Scientist, PIEAS | abdulmajiid@pieas.edu.pk